



COMP4121 Advanced Algorithms

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More randomised algorithms: Gaussian Annulus, Random Projection and Johnson
Lindenstrauss Lemmas

Details to be found on pages 12-27 of the Blum, Hopcroft and Kannan textbook

The Curse of High Dimensionality

- Assume you have a large collection $C = \{D_i : 1 \leq i \leq 1,000,000\}$ of 1,000,000 documents and you get a new document D .
- You want to find a few documents in your database which are the most similar to document D .
- One simple way of comparing documents is to use a large dictionary of, say 100,000 words and for each document find the relative frequency of each word from your dictionary in every document.
- In this way every document is represented by a vector of dimension 100,000 where each coordinate i represents the frequency of i^{th} word from your dictionary.
- You now want to find documents from C which are closest to document D in the usual Euclidean metric.
- But looking for nearest neighbours in a space of dimensionality 100,000 is extremely slow!

The Curse of High Dimensionality

- We are looking at finding k such that

$$d(D_k, D) = \sqrt{\sum_{j=1}^{100,000} ((D_k)_j - D_j)^2}$$

is minimised.

- Another related task might be to cluster all documents in C into a small number of clusters of documents which are in close proximity.
- Clearly, both finding nearest neighbours and clustering in high dimensional spaces are difficult tasks.
- What can we do in such circumstances to make our algorithms faster?

The Curse of High Dimensionality

- Choose a much smaller collection of vectors such that if we project our vectors onto the subspace spanned by such vectors the distances between points do not change much, i.e., points which are nearby have nearby projections and points which are distant have distant projections.
- How do we find such vectors??
- Choose them randomly and hope for the best!
- This is why we are interested in random projections in high dimensional spaces.

Generating random points in d -dimensional spaces $\mathbb{R}^d$

- Let us consider a Gaussian random variable X with zero mean ($E[X] = \mu = 0$) and variance $V[X] = v = 1$; its density is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi v}} e^{-\frac{x^2}{2v}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

- Assume that we use such X to generate independently the coordinates $(x_1, \dots, x_d)$ of a random vector $\vec{x}$ from $\mathbb{R}^d$.
- Since $E[X] = 0$ we have $E[X^2] = E[(X - E[X])^2] = V[X] = 1$.
- Thus, also $E\left[\frac{X_1^2 + \dots + X_d^2}{d}\right] = dV[X]/d = V[X]$.
- We denote by $|x|$ the norm (here just the length) of a vector x ,
 $|x| = \sqrt{x_1^2 + \dots + x_d^2}$,
- We now have $E\left[\frac{|x|^2}{d}\right] = V[X] = 1$.
- Thus the expected value of the square of the length of such a random vector is $E[|x|^2] = d$.
- So, on average, $|x| \approx \sqrt{d}$.

Generating random points in d -dimensional spaces $\mathbb{R}^d$

- Also, by the Law of Large Numbers,

$$\mathcal{P} \left(\left| \frac{x_1^2 + \dots + x_d^2}{d} - V[X] \right| > \varepsilon \right) \leq \frac{V[X^2]}{d\varepsilon^2}$$

- Thus,

$$\mathcal{P} \left(\left| \frac{|x|^2}{d} - 1 \right| > \varepsilon \right) \leq \frac{V[X^2]}{d\varepsilon^2} \quad \text{i.e.,} \quad \mathcal{P} \left(\left| \frac{|x|^2}{d} - 1 \right| \leq \varepsilon \right) \geq 1 - \frac{V[X^2]}{d\varepsilon^2}$$

- This implies that, if d is large, then with high probability

$$\frac{|x|^2}{d} \approx 1 \quad \Rightarrow \quad |x| \approx \sqrt{d} = \Theta(\sqrt{d})$$

i.e., thus generated point is at a distance $\Theta(\sqrt{d})$ from the origin.

- If we chose 2 points independently, then

$$E[\langle x, y \rangle] = E[x_1y_1 + \dots + x_dy_d] = d E[XY] = d E[X]E[Y] = 0$$

- This means that the expected value of the scalar product of any two vectors with independently randomly chosen coordinates is zero.
- So, intuitively, vectors with randomly chosen coordinates have approximately the same length $\Theta(\sqrt{d})$ and any two such random vectors are likely to be almost orthogonal!**

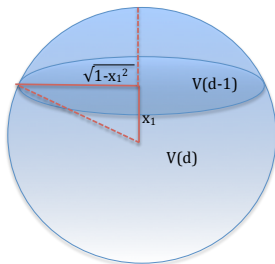
High dimensional spaces are quite counter-intuitive!

- The above should hint that high dimensional spaces are quite counter-intuitive.
- We have to be careful not to transfer our 3D intuition to high dimensional spaces.
- We now show two strange facts about high dimensional balls:
 - ① Most of the volume of a high dimensional ball is near any of its equators. By this we mean that if we cut a high dimensional ball with any two parallel hyper-planes symmetric with respect to the center and close to the center, then most of the volume of the ball is between these two hyper planes.
 - ② Most of the volume of a high dimensional ball is near its surface. By this we mean that if we consider a ball of radius r and a slightly smaller ball of radius $r(1 - \epsilon)$ then most of the volume of the bigger ball is in the annulus outside the smaller ball.
- These facts are not just curiosities, but have importance for algorithms we will study.

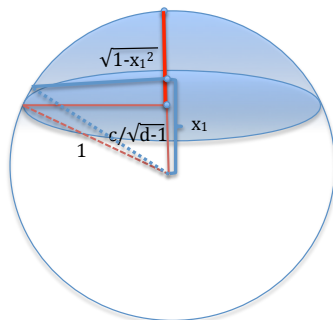
Generating random points in d -dimensional spaces $\mathbb{R}^d$

- We first note that the volume of a sphere of radius r is proportional to r^d , in the sense that, if we denote by $V(d)$ the volume of the d -dimensional ball of radius 1, then the volume of a d dimensional ball of radius r is equal to $r^d V(d)$. (This is clear for the d dimensional cube and it follows for any other solid by the exhaustion principle: we approximate the solid by a union of small disjoint hypercubes).
- This implies that the volume of a d -dimensional ball can be represented as the following integral:

$$V(d) = \int_{-1}^1 \left(\sqrt{1 - x_1^2} \right)^{d-1} V(d-1) dx_1$$



Generating random points in d -dimensional spaces $\mathbb{R}^d$



- Let c be a constant. Consider the portion S of a unit ball consisting of all points such that their x_1 coordinate satisfies $x_1 \geq \frac{c}{\sqrt{d-1}}$.
- Let A be the volume of such a slice S .
- Then

$$A = \int_{\frac{c}{\sqrt{d-1}}}^1 \left(\sqrt{1-x_1^2} \right)^{d-1} V(d-1) dx_1$$

- (Recall that our goal is to show that two randomly and independently generated vectors are almost certainly almost orthogonal; to achieve this we need to perform some mumbo-jumbo whose purpose will be clear a bit later).

Generating random points in d -dimensional spaces $\mathbb{R}^d$

- We need the inequality $1 - x \leq e^{-x}$ for all $0 \leq x \leq 1$.
- For a proof, see the refresher notes on probability and statistics, available at the course web site.
- Using such an inequality, from what we have

$$A = \int_{\frac{c}{\sqrt{d-1}}}^1 \left(\sqrt{1 - x_1^2} \right)^{d-1} V(d-1) dx_1$$

we obtain

$$A < \int_{\frac{c}{\sqrt{d-1}}}^1 (e^{-x_1^2})^{\frac{d-1}{2}} V(d-1) dx_1$$

Over the slice S we have $x_1 \geq \frac{c}{\sqrt{d-1}}$ which implies $\frac{x_1 \sqrt{d-1}}{c} \geq 1$ and this implies

$$A < \int_{\frac{c}{\sqrt{d-1}}}^1 \frac{x_1 \sqrt{d-1}}{c} e^{-x_1^2 \frac{d-1}{2}} V(d-1) dx_1$$

- The purpose of adding this term is to allow substitution of variable x_1 with variable $u = x_1^2$; then $du = 2x_1 dx_1$.

Generating random points in d -dimensional spaces $\mathbb{R}^d$

- Thus, with $u = x_1^2$ we have $du = 2x_1 dx_1$ and so, from

$$A < \int_{\frac{c}{\sqrt{d-1}}}^1 \frac{x_1 \sqrt{d-1}}{c} e^{-x_1^2 \frac{d-1}{2}} V(d-1) dx_1$$

we obtain

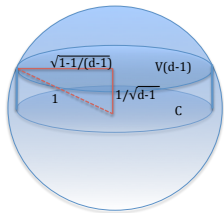
$$\begin{aligned} A &< \int_{\frac{c^2}{d-1}}^1 \frac{\sqrt{d-1}}{c} e^{-u \frac{d-1}{2}} V(d-1) \frac{1}{2} du \\ &< \frac{\sqrt{d-1}}{2c} V(d-1) \int_{\frac{c^2}{d-1}}^{\infty} e^{-u \frac{d-1}{2}} du \\ &= -\frac{\sqrt{d-1}}{2c} V(d-1) \frac{1}{\frac{d-1}{2}} e^{-u \frac{d-1}{2}} \Big|_{u=\frac{c^2}{d-1}}^{u=\infty} \\ &= \frac{\sqrt{d-1}}{2c} V(d-1) \frac{1}{\frac{d-1}{2}} e^{-\frac{d-1}{2} \frac{c^2}{d-1}} \\ &= \frac{1}{c\sqrt{d-1}} V(d-1) e^{-\frac{c^2}{2}} \end{aligned}$$

- So we have obtained that the volume A of the slice S of the unit ball which lies above the hyperplane $x_1 = \frac{c}{\sqrt{d-1}}$ satisfies

$$A < V(d-1) \frac{e^{-\frac{c^2}{2}}}{c\sqrt{d-1}}$$

- We now want to show that such a volume is small compared to half the volume of the entire ball. However, we do not want to use $V(d)$, we want an expression involving $V(d-1)$ which can cancel $V(d-1)$ in the above bound.
- Note that the volume of the whole hemisphere H is larger than the volume of a cylinder below the intersection of the ball and the plane $x_1 = \frac{1}{\sqrt{d-1}}$, which has the volume C which is the product of the $d-1$ dimensional “surface area” of its base which is equal to $V(d-1) \left(\sqrt{1 - \frac{1}{d-1}} \right)^{d-1}$ and its height which is $1/\sqrt{d-1}$.
- Thus, the volume C of such a cylinder is

$$\begin{aligned} C &= V(d-1) \left(\sqrt{1 - \frac{1}{d-1}} \right)^{d-1} \frac{1}{\sqrt{d-1}} \\ &= V(d-1) \left(1 - \frac{1}{d-1} \right)^{\frac{d-1}{2}} \frac{1}{\sqrt{d-1}} \end{aligned}$$



- We now need yet another inequality: for all $\alpha \geq 1$ and all $0 < x < 1$,

$$(1 - x)^\alpha \geq 1 - \alpha x$$

(This inequality can be easily proved by considering $\beta(x) = (1 - x)^\alpha - 1 + \alpha x$; again, $\beta(0) = 0$ and $\beta'(x) = -\alpha(1 - x)^{\alpha-1} + \alpha = \alpha(1 - (1 - x)^{\alpha-1}) \geq 0$ which implies that $\beta(x)$ is non-decreasing and thus non-negative.)

- Applying the above inequality with $\alpha = (d - 1)/2$ and $x = 1/(d - 1)$ we obtain

$$\begin{aligned} C &= V(d - 1) \left(1 - \frac{1}{d - 1}\right)^{\frac{d-1}{2}} \frac{1}{\sqrt{d - 1}} \\ &\geq V(d - 1) \left(1 - \frac{1}{d - 1} \frac{d - 1}{2}\right) \frac{1}{\sqrt{d - 1}} \\ &\geq V(d - 1) \frac{1}{2\sqrt{d - 1}} \end{aligned}$$

- So we got that the volume A of the slice of the unit ball which is above the plane $x_1 = \frac{c}{\sqrt{d-1}}$ and the volume H of the whole hemisphere satisfy

$$A < V(d-1) \frac{e^{-\frac{c^2}{2}}}{c\sqrt{d-1}}$$

$$H \geq C \geq V(d-1) \frac{1}{2\sqrt{d-1}}$$

- Thus, the ratio of the volume of the slice of the sphere and the whole hemisphere satisfies

$$\frac{A}{H} < \frac{V(d-1) \frac{e^{-\frac{c^2}{2}}}{c\sqrt{d-1}}}{V(d-1) \frac{1}{2\sqrt{d-1}}} = \frac{2}{c} e^{-\frac{c^2}{2}}$$

- **Corollary:** If we pick uniformly at random a point from the unit ball, the probability that its coordinate x_1 satisfies $|x_1| > \frac{c}{\sqrt{d-1}}$ is less than $\frac{2}{c} e^{-\frac{c^2}{2}}$.
- **This also means that most of the volume of the unit ball is between the hyperplanes $x_1 = -\frac{c}{\sqrt{d-1}}$ and $x_1 = \frac{c}{\sqrt{d-1}}$, i.e., near the equator.**
- Also, the annulus consisting of all x such that $r \leq |x| \leq 1$ for $r < 1$ has volume $V(d) - r^d V(d) = V(d)(1 - r^d)$ so **if d is large almost all of the volume of the unit ball is near its surface.**

- If we randomly and independently choose 2 points x, x' from the unit ball, we can rotate the ball so that x is in the direction of x_1 coordinate.
- Thus, the above two facts guarantee that **with high probability** $|x| \approx 1$ and $|x'| \approx 1$ **as well as that the projection of x' onto x has, with high probability** $\geq 1 - \frac{2}{c}e^{-\frac{c^2}{2}}$, **a value smaller than** $\frac{c}{\sqrt{d-1}}$, **i.e., that x and x' are almost orthogonal.**
- Next theorem generalises this to n points.
- It shows that if we draw randomly a relatively small number n of vectors compared to the dimension d , we are almost guaranteed that they will be of approximately the same length and that form an almost orthogonal basis.
- They will span a space onto which we will project vectors from $\mathbb{R}^d$ into a space of dimension n .
- The purpose is to reduce the dimensionality of the space.
- **Theorem:** Assume we draw independently and uniformly n points $x_1, \dots, x_n$ from the unit ball.
Then with probability $> 1 - \frac{3}{2} \frac{1}{n}$ for all $1 \leq i, j \leq n, i \neq j$,
 - 1 $1 \geq |x_i| > 1 - \frac{2 \ln n}{d}$;
 - 2 $|\langle x_i, x_j \rangle| \leq \sqrt{\frac{6 \ln n}{d-1}}$.

- **Proof:** Probability that $|x_i| < 1 - \varepsilon$ is

$$\frac{V(d)(1 - \varepsilon)^d}{V(d)} = (1 - \varepsilon)^d < e^{-\varepsilon d}$$

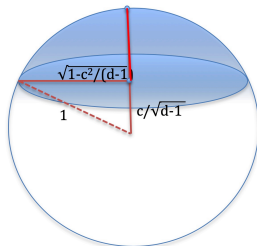
- Thus, taking $\varepsilon = \frac{2 \ln n}{d}$ we obtain that for each i

$$\mathcal{P} \left(|x_i| < 1 - \frac{2 \ln n}{d} \right) < e^{-\frac{2 \ln n}{d} d} = e^{-\ln n^2} = \frac{1}{n^2}$$

- Thus, the probability that $|x_i| < 1 - \frac{2 \ln n}{d}$ for at least one i is less than $n \cdot \frac{1}{n^2} = \frac{1}{n}$. This reasoning is called the *Union bound*.
- *Union bound:* if you have n sets $A_1, \dots, A_n$ (not necessarily disjoint) such that $\mathcal{P}(x \in A_i) = p_i$ then $\mathcal{P}(x \in \cup_{i=1}^n A_i) \leq \sum_{i=1}^n p_i$.



- To prove the second claim we rotate the sphere so that x_j is the polar axis.



- We now note that, since $|x_i|, |x_j| \leq 1$,

$$|\langle x_i, x_j \rangle| = |x_i| |x_j| |\cos \angle(x_i, x_j)| \leq |x_i| |\cos \angle(x_i, x_j)|$$

- Thus, $|\langle x_i, x_j \rangle| \geq \frac{\sqrt{6 \ln n}}{\sqrt{d-1}}$ implies that the length of the orthogonal projection of x_i onto x_j , i.e., $|x_i| |\cos \angle(x_i, x_j)|$ is larger than $\frac{\sqrt{6 \ln n}}{\sqrt{d-1}}$.
- But we have seen in the previous corollary that if we pick a point x_i uniformly at random a point from the unit ball, the probability that its coordinate x_j satisfies $|x_j| > \frac{c}{\sqrt{d-1}}$ is less than $\frac{2}{c} e^{-\frac{c^2}{2}}$.
- Letting $c = \sqrt{6 \ln n}$, we obtain that for every pair i, j

$$\mathcal{P} \left(|\langle x_i, x_j \rangle| \geq \frac{\sqrt{6 \ln n}}{\sqrt{d-1}} \right) \leq \frac{2}{\sqrt{6 \ln n}} e^{-\frac{6 \ln n}{2}} = \frac{2}{\sqrt{6 \ln n}} e^{-\ln n^3}$$

- Simplifying the expression $\frac{2}{\sqrt{6 \ln n}} e^{-\ln n^3}$ we obtain

$$\mathcal{P} \left(|\langle x_i, x_j \rangle| \geq \frac{\sqrt{6 \ln n}}{\sqrt{d-1}} \right) \leq \frac{2}{\sqrt{6 \ln n}} \frac{1}{n^3} < \frac{1}{n^3}.$$

- There are $\binom{n}{2}$ pairs, so again by union bound, the probability that for at least one pair x_i, x_j the scalar product is large, i.e., $|\langle x_i, x_j \rangle| \geq \frac{\sqrt{6 \ln n}}{\sqrt{d-1}}$ is at most $\binom{n}{2} \frac{1}{n^3} = \frac{n(n-1)}{2} \frac{1}{n^3} < \frac{1}{2n}$.
- Thus, all inequalities of both kinds hold with a probability $(1 - \frac{1}{n})(1 - \frac{1}{2n}) = 1 - \frac{1}{n} - \frac{1}{2n} + \frac{1}{2n^2} > 1 - \frac{3}{2} \frac{1}{n}$.
- Thus, to summarise, we have proved that, if we draw independently and uniformly n points $x_1, \dots, x_n$ from the unit ball, then with probability $1 - \frac{3}{2} \frac{1}{n}$ for all $1 \leq i, j \leq n, i \neq j$,
 - ❶ $|x_i| > 1 - \frac{2 \ln n}{d}$;
 - ❷ $|\langle x_i, x_j \rangle| \leq \sqrt{\frac{6 \ln n}{d-1}}$.
- But how do we draw random points from a unit ball?
- It turns out that it is easier to draw points using a spherical Gaussian which is what we do next.

Gaussian Annulus Theorem

- We need the following lemma which we do not prove; its proof is based on tail inequalities which can be found in the BHK book. Recall that for $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ we let $|x| = \sqrt{x_1^2 + \dots + x_d^2}$.
- **Lemma:** Let X be a d -dimensional spherical Gaussian with zero mean and unit variance in each direction, i.e., the with the probability density

$$f_X(x) = \prod_{i=1}^d \frac{1}{\sqrt{2\pi}} e^{-\frac{x_i^2}{2}} = \frac{1}{(2\pi)^{\frac{d}{2}}} e^{-\frac{\sum x_i^2}{2}} = \frac{1}{(2\pi)^{\frac{d}{2}}} e^{-\frac{|x|^2}{2}}$$

If we draw from such a distribution a random vector x , then the probability that x belongs to the annulus (hollow ball of outer radius $\sqrt{d} + \beta$ with the shell of thickness 2β)

$$\{x : \sqrt{d} - \beta \leq |x| \leq \sqrt{d} + \beta\} = \{x : \left| |x| - \sqrt{d} \right| < \beta\}$$

is at least $1 - 3e^{-\frac{\beta^2}{96}}$.

- (For convenience of notation we denote $1/96$ by γ , so the probability that x belongs to the annulus is at least $1 - 3e^{-\gamma\beta^2}$).

- Let us choose k random vectors $u_1, \dots, u_k \in \mathbb{R}^d$ by sampling randomly a spherical Gaussian with a unit variance (i.e., we generate k vectors with each of d many coordinates obtained by sampling randomly and independently the one dimensional Gaussian).
- For any vector v define $f(v)$, called *the random projection*, by

$$f(v) = (\langle v, u_1 \rangle, \dots, \langle v, u_k \rangle)$$

- Theorem: (Random Projection Lemma)**

$$\mathcal{P} \left(\left| |f(v)| - \sqrt{k}|v| \right| \geq \varepsilon \sqrt{k}|v| \right) \leq 3e^{-\gamma k \varepsilon^2}$$

- Proof:** Since $f(v)$ is a linear operator in v , we can divide both sides of the inner inequality with $|v|$. Thus, we can assume that $|v| = 1$ and we have to show that for $|v| = 1$

$$\mathcal{P} \left(\left| |f(v)| - \sqrt{k} \right| \geq \varepsilon \sqrt{k} \right) \leq 3e^{-\gamma k \varepsilon^2}$$

- Note that $\langle u_i, v \rangle = \sum_{j=1}^d u_{i,j} v_j$, since $u_{i,j}$ are Gaussians, so is their linear combination; thus, $\langle u_i, v \rangle$ is also Gaussian.
- Since $u_{i,j}$ are of unit variance and independent, we obtain

$$V[\langle u_i, v \rangle] = V \left[\sum_{j=1}^d u_{i,j} v_j \right] = \sum_{j=1}^d V[u_{i,j}] v_j^2 = \sum_{j=1}^d v_j^2 = |v| = 1.$$

- Thus, $\langle u_i, v \rangle$ is also Gaussian of unit variance.
- It follows that the theorem is a direct consequence of the Gaussian Annulus Theorem with $\beta = \varepsilon \sqrt{k}$.

- Let us define $f^*(v) = f(v)/\sqrt{k}$;
- if we divide by $\sqrt{k}$ both sides of the inner inequality in

$$\mathcal{P}\left(\left||f(v)| - \sqrt{k}|v|\right| \geq \varepsilon\sqrt{k}|v|\right) \leq 3e^{-\gamma k \varepsilon^2}$$

we obtain the following consequence:

- **Corollary:**

$$\mathcal{P}\left(\left||f^*(v)| - |v|\right| \geq \varepsilon|v|\right) \leq 3e^{-\gamma k \varepsilon^2}$$

- Thus, the modified projection mapping $f^*(v)$ “almost preserves” length of vectors.
- This corollary is needed to prove the following important theorem with lots of practical applications.

Johnson-Lindenstrauss Lemma

- **Theorem: (Johnson-Lindenstrauss Lemma)** For any ε , $0 < \varepsilon < 1$, and any integer n , assume that k satisfies $k \geq \frac{3}{\gamma \varepsilon^2} \ln n$. Then for any set of n points given by vectors $v_1, \dots, v_n$ in $\mathbb{R}^d$, with the probability of at least $1 - \frac{3}{2n}$, the random projection $f^*: \mathbb{R}^d \rightarrow \mathbb{R}^k$ has the property that for ALL pairs of points v_i, v_j

$$||f^*(v_i - v_j)| - |v_i - v_j|| \leq \varepsilon |v_i - v_j|$$

Thus, $f^*(v)$ “almost” preserves distances between points given by vectors v_i , despite reduction of dimensionality from $d \gg k$ to only k .

- **Proof:** By applying the Random Projection Lemma, we get that for any pair v_i, v_j ,

$$\begin{aligned} \mathcal{P} (||f^*(v_i - v_j)| - |v_i - v_j|| \geq \varepsilon |v_i - v_j|) &\leq 3e^{-\gamma k \varepsilon^2} \leq 3e^{-\gamma \frac{3}{\gamma \varepsilon^2} \ln n \varepsilon^2} \\ &\leq 3e^{-\ln n^3} = \frac{3}{n^3} \end{aligned}$$

- Since there are $\binom{n}{2} < \frac{n^2}{2}$ pairs, the probability that at least one of the above inequalities fails is less than $\frac{n^2}{2} \cdot \frac{3}{n^3} = 1 - \frac{3}{2n}$. (Union bound again).
- Note that k is a linear function of $\ln n$, so it grows slowly, allowing to handle efficiently lots of points v_i .

Johnson-Lindenstrauss Lemma

- Main application of the Johnson Lindenstrauss Lemma: nearest neighbour search in spaces of extremely high dimension, such as collections of documents, represented by the vector of the relative frequencies of each word from a dictionary (of key words, for example).
- Given a database of a large number of papers and a new paper p , find a paper p_i from the database which is the “most similar” to p , in the sense that the same keywords appear with approximately same frequencies.
- The dictionary of all keywords is very large, in thousands or even tens of thousands, so nearest neighbour search would be extremely slow.
- Even if your database had 100,000 entries, $\ln(100,000) < 12$

Applications of the Johnson-Lindenstrauss Lemma

- Solution: choose a large enough k , i.e., $k \geq \frac{288}{\varepsilon^2} \ln(|DB| + |Q|)$ where $|DB|$ is the size of the database and $|Q|$ the expected number of queries to be made.
- Recall that $\gamma = 1/96$ so $3/\gamma = 288$. However, with more careful analysis, the requirement $k \geq \frac{288}{\varepsilon^2} \ln n$ can be reduced to $k \geq \frac{24}{3\varepsilon^2 - 2\varepsilon^3} \ln n$.
- A good dictionary contains more than 100,000 words.
- So if you want to compare English documents according to the relative frequency of words, each document has to be represented as a vector in 100,000 dimensional space!
- Assume that your database has 100,000 documents stored; you want to find the most similar documents in your database for a batch of 10,000 new documents. How many random vectors should you choose to project onto?
- If we set $\varepsilon = 0.1$ we obtain

$$k \geq \frac{24}{3\varepsilon^2 - 2\varepsilon^3} \ln(|DB| + |Q|) = \frac{24}{3 \times .01 - 2 \times .001} \times \ln(110,000) = 9950,$$

so more than 10-folds reduction of dimensionality.

Applications of the Johnson-Lindenstrauss Lemma

- So we choose k random vectors, by choosing each coordinate of every vector using a unit variance Gaussian.
- Pre-process your database by replacing each vector $x_j \in DB$, ($x_j \in \mathbb{R}^d$) with its scaled projection $f^*(x_j) = f(x_j)/\sqrt{k}$ to the $k \ll d$ dimensional subspace spanned by these k random vectors.
- As each query y arrives, also replace $y \in \mathbb{R}^d$ with its projection $f^*(y) = f(y)/\sqrt{k}$.
- Do the search for the nearest neighbour of $f(y)/\sqrt{k}$ in the projected k dimensional space instead of the search for the nearest neighbour of y in the space of dimension $d \gg k$.